

Disturbance-Observer-Enhanced Backstepping Super-Twisting Control for Robust Regulation of a Nonlinear Quadruple-Tank System

Wytze Hempenius

September 2026

Abstract

This paper proposes a disturbance-observer-enhanced backstepping super-twisting controller (DO-BSSTC) for robust liquid-level regulation of a nonlinear quadruple-tank system (QTS). The proposed method extends a fixed-gain backstepping super-twisting controller (BSSTC) by introducing online estimation and feed-forward compensation of an unknown disturbance acting on the plant. The objective is to preserve the smooth actuator behavior of the super-twisting controller while improving rejection of unmodeled pump-flow disturbances.

A nonlinear QTS model is simulated without linearization around a single operating point. Three controllers are compared: conventional first-order sliding mode control (SMC), fixed-gain BSSTC, and the proposed DO-BSSTC. Performance is evaluated using the integral absolute error (IAE), integral squared error (ISE), integral time absolute error (ITAE), control total variation (TV), and RMS control effort.

Under nominal conditions, the DO-BSSTC produces nearly identical tracking performance to the fixed-gain BSSTC, indicating that the disturbance observer does not significantly modify the nominal closed-loop behavior. For a pump-flow disturbance of $-300 \text{ cm}^3/\text{s}$ applied between 700 and 900 s, however, the proposed controller substantially improves disturbance rejection. The tank-4 disturbance-window IAE decreases from 27.56 for BSSTC to 0.40 for DO-BSSTC, corresponding to a 98.55% reduction. The mean absolute tank-4 error decreases from 0.1378 cm to 0.0020 cm. Furthermore, the measured post-disturbance recovery time using the criterion $|e_4| < 0.1 \text{ cm}$ decreases from 0.11 s to 0.01 s.

The results demonstrate that explicit disturbance estimation can substantially improve the disturbance-rejection capability of a continuous super-twisting controller while preserving its low control variation under nominal operation.

Keywords: Quadruple-tank system; nonlinear control; sliding mode control; backstepping; super-twisting control; disturbance observer; robust control; liquid-level control.

1 Introduction

The control of nonlinear multivariable processes remains an important problem in industrial automation because nonlinearities, actuator constraints, plant interactions, and external disturbances can significantly affect closed-loop performance. The quadruple-tank system (QTS) is a widely used benchmark for investigating nonlinear and multivariable control because it combines nonlinear hydraulic dynamics, multiple interacting states, and multiple manipulated inputs.

A QTS consists of four interconnected liquid tanks driven by two pumps. The nonlinear behavior originates primarily from the square-root relationship between liquid height and outlet flow. Consequently, a controller designed from a linearized model can provide good performance around a particular operating point while losing accuracy when the operating conditions change.

Sliding mode control (SMC) is attractive for nonlinear systems because of its robustness to model uncertainty and disturbances. However, conventional first-order SMC commonly generates high-frequency switching in the control signal. This phenomenon, known as chattering, can be undesirable in practical applications because it may cause excessive actuator activity and mechanical wear.

Backstepping combined with super-twisting control provides an alternative approach. Aranda-Cetraro et al. proposed a modulating-function-based backstepping super-twisting controller for the QTS and demonstrated that the approach can provide robust nonlinear liquid-level regulation while substantially reducing the chattering associated with first-order sliding mode control [1]. Their work also investigated the use of modulating functions to avoid direct analytical differentiation and to increase robustness against measurement noise.

The present work builds on this control structure and focuses on a different limitation: explicit rejection of unknown disturbances. A standard fixed-gain BSSTC can reject disturbances through its robust feedback action, but it does not explicitly estimate the magnitude of the disturbance. Therefore, the controller must generate a corrective response after the disturbance affects the plant.

To address this limitation, this paper proposes a *disturbance-observer-enhanced backstepping super-twisting controller* (DO-BSSTC). The main idea is to estimate the unknown disturbance from the difference between the measured plant dynamics and the nominal nonlinear model. The estimated disturbance is then converted into a feed-forward pump compensation.

The proposed approach therefore combines two complementary mechanisms: the super-twisting controller provides robust continuous feedback control, while the disturbance observer provides explicit compensation of the estimated disturbance.

The main contributions of this work are:

1. Development of a disturbance-observer extension of a backstepping super-twisting controller for the nonlinear QTS.
2. Direct comparison between first-order SMC, fixed-gain BSSTC, and the proposed DO-BSSTC.
3. Quantitative evaluation using IAE, ISE, ITAE, control total variation, and RMS control effort.
4. Evaluation of disturbance rejection using a $-300 \text{ cm}^3/\text{s}$ pump-flow disturbance.
5. Demonstration that the proposed disturbance compensation can reduce the disturbance-window tank-4 IAE by 98.55% compared with the fixed-gain BSSTC in the investigated simulation.

The remainder of this paper is organized as follows. Section II presents the nonlinear QTS model. Section III describes the baseline controllers. Section IV develops the proposed disturbance-observer-enhanced BSSTC. Section V presents the simulation methodology and performance metrics. Section VI discusses the simulation results. Finally, Section VII presents the conclusions and directions for future work.

2 Nonlinear Quadruple-Tank System Model

The nonlinear QTS model used in this study is based on the mass-balance and Bernoulli-law formulation commonly used for quadruple-tank systems [1].

The state vector is defined as

$$\mathbf{x} = \begin{bmatrix} h_1 & h_2 & h_3 & h_4 \end{bmatrix}^T, \quad (1)$$

where h_i denotes the liquid level of tank i . The manipulated-input vector is

$$\mathbf{u} = \begin{bmatrix} u_1 & u_2 \end{bmatrix}^T, \quad (2)$$

where u_1 and u_2 are the pump flow commands.

The nonlinear plant is represented by

$$\dot{h}_1 = -\frac{a_1\sqrt{2gh_1}}{A_1} + \frac{(1-\gamma_2)k_2u_2}{A_1} + \xi_1, \quad (3)$$

$$\dot{h}_2 = -\frac{a_2\sqrt{2gh_2}}{A_2} + \frac{(1-\gamma_1)k_1u_1}{A_2} + \xi_2, \quad (4)$$

$$\dot{h}_3 = -\frac{a_3\sqrt{2gh_3}}{A_3} + \frac{a_2\sqrt{2gh_2}}{A_3} + \frac{\gamma_2k_2u_2}{A_3} + \xi_3, \quad (5)$$

and

$$\dot{h}_4 = -\frac{a_4\sqrt{2gh_4}}{A_4} + \frac{a_1\sqrt{2gh_1}}{A_4} + \frac{\gamma_1k_1u_1}{A_4} + \xi_4. \quad (6)$$

The parameters used in the simulations are

Table 1: Quadruple-tank system parameters.

Parameter	Value
A_1, A_2, A_3, A_4	706.85 cm ²
a_1, a_2	1.89 cm ²
a_3, a_4	5.39 cm ²
g	981 cm/s ²
γ_1	0.80
γ_2	0.90
k_1, k_2	1.00
$U_{\max}$	2863 cm ³ /s
U_{LO}	0.10 $U_{\max}$
U_{HI}	0.60 $U_{\max}$

The manipulated inputs are constrained according to

$$U_{\text{LO}} \leq u_i \leq U_{\text{HI}}. \quad (7)$$

The physical liquid-level limits are

$$0 \leq h_i \leq 45 \text{ cm}. \quad (8)$$

The controlled outputs are the liquid levels of tanks 3 and 4. The reference trajectories are piecewise constant:

$$(h_3^*, h_4^*) = \begin{cases} (40, 7), & 0 \leq t < 300, \\ (10, 34), & 300 \leq t < 600, \\ (24, 15), & 600 \leq t \leq 1000. \end{cases} \quad (9)$$

Thus, the control objective is

$$h_3(t) \rightarrow h_3^*(t), \quad h_4(t) \rightarrow h_4^*(t). \quad (10)$$

3 Baseline Control Strategies

3.1 First-Order Sliding Mode Controller

The first baseline controller is a conventional first-order SMC. The sliding variables are defined as

$$\sigma_1 = h_4 - h_4^*, \quad (11)$$

$$\sigma_2 = h_3 - h_3^*. \quad (12)$$

The control law uses a saturated relay:

$$u_1 = u_{\text{mid}} - u_{\text{amp}} \text{sign}(\sigma_1), \quad (13)$$

$$u_2 = u_{\text{mid}} - u_{\text{amp}} \text{sign}(\sigma_2), \quad (14)$$

where

$$u_{\text{mid}} = \frac{U_{\text{HI}} + U_{\text{LO}}}{2}, \quad (15)$$

and

$$u_{\text{amp}} = \frac{U_{\text{HI}} - U_{\text{LO}}}{2}. \quad (16)$$

Although the relay structure provides robust tracking, rapid switching can generate substantial actuator activity.

3.2 Fixed-Gain BSSTC

The second controller is a two-step backstepping super-twisting controller. The nonlinear tank dynamics are transformed using the square-root variables

$$x_1 = \sqrt{h_{\text{out}}}, \quad x_2 = \sqrt{h_{\text{in}}}. \quad (17)$$

For each controlled tank subsystem, the dynamics can be written in the form

$$\dot{x}_1 = Ax_1 + Bx_2 + Cu. \quad (18)$$

A virtual control α_1 is constructed through the backstepping procedure. The second backstepping error is

$$z_2 = x_2 - \alpha_1. \quad (19)$$

A super-twisting reaching law is then used:

$$\dot{s} = -c_2 s - \lambda_1 |s|^{1/2} \text{sign}(s) - v, \quad (20)$$

with

$$\dot{v} = \lambda_2 \text{sign}(s). \quad (21)$$

The pump command is obtained by integrating the control derivative $w = \dot{u}$. This provides a continuous actuator command and avoids the direct switching behavior of first-order SMC.

4 Proposed Disturbance-Observer-Enhanced BSSTC

4.1 Motivation

The fixed-gain BSSTC provides robust feedback control but does not explicitly reconstruct the unknown disturbance. When a disturbance affects the plant, the feedback controller first observes the resulting tracking error and then generates corrective action.

The proposed method introduces a disturbance observer that estimates the unknown disturbance directly from the plant dynamics.

The approach is particularly suitable for the QTS because the nonlinear model provides an explicit expression for the expected tank-level derivative.

4.2 Disturbance Estimation

For tank 4, Equation (6) can be written as

$$\dot{h}_4 = f_4(\mathbf{x}, u_1) + \xi_4, \quad (22)$$

where

$$f_4(\mathbf{x}, u_1) = -\frac{a_4\sqrt{2gh_4}}{A_4} + \frac{a_1\sqrt{2gh_1}}{A_4} + \frac{\gamma_1 k_1}{A_4} u_1. \quad (23)$$

The unknown disturbance can therefore be estimated from

$$\hat{\xi}_4 = \dot{h}_4^{\text{meas}} - f_4(\mathbf{x}, u_1). \quad (24)$$

To avoid excessive sensitivity to derivative estimation errors, a first-order filtered estimate is used:

$$\tau_d \dot{\hat{\xi}}_4 + \hat{\xi}_4 = \xi_{4,\text{raw}}, \quad (25)$$

where τ_d is the observer filtering parameter.

This observer provides an estimate of the disturbance acting on the tank-4 subsystem without requiring the disturbance itself to be known in advance.

4.3 Feed-Forward Compensation

The input gain associated with u_1 in the tank-4 equation is

$$C_4 = \frac{\gamma_1 k_1}{A_4}. \quad (26)$$

The disturbance compensation is therefore defined as

$$u_{\text{ff}} = -\frac{\hat{\xi}_4}{C_4}. \quad (27)$$

The proposed pump command can conceptually be expressed as

$$u_1 = u_{\text{BSSTC}} + u_{\text{ff}}, \quad (28)$$

subject to the physical actuator constraints

$$u_1 = \text{sat}(u_{\text{BSSTC}} + u_{\text{ff}}). \quad (29)$$

The complete control architecture is therefore composed of two parallel mechanisms:

1. BSSTC feedback control for nonlinear robust stabilization.

2. Disturbance-observer feed-forward compensation for rejection of the estimated unknown disturbance.

This architecture is illustrated conceptually in Fig. 1.

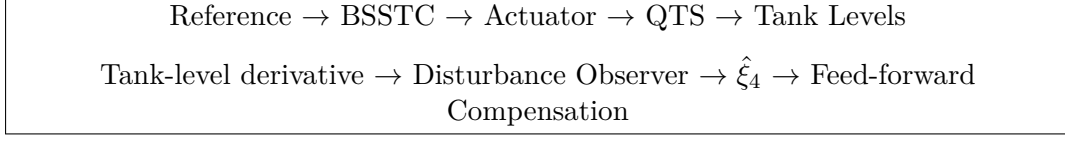


Figure 1: Conceptual architecture of the proposed DO-BSSTC.

5 Simulation Methodology

The nonlinear QTS was simulated directly from the nonlinear differential equations without linearization around a fixed operating point.

The numerical integration was performed using a fourth-order Runge–Kutta method with a fixed simulation step of

$$T_s = 0.01 \text{ s.} \quad (30)$$

The total simulation time was

$$T = 1000 \text{ s.} \quad (31)$$

The initial liquid levels were

$$h_1(0) = h_2(0) = h_3(0) = h_4(0) = 5 \text{ cm.} \quad (32)$$

Three controllers were evaluated:

1. First-order SMC.
2. Fixed-gain BSSTC.
3. Proposed DO-BSSTC.

The controllers were evaluated under both nominal conditions and disturbed conditions.

5.1 Disturbance Scenario

To evaluate robustness, an unknown pump-flow loss was introduced into the tank-4 dynamics:

$$f_{p1} = -300 \text{ cm}^3/\text{s.} \quad (33)$$

The disturbance was applied during

$$700 \leq t \leq 900 \text{ s.} \quad (34)$$

The corresponding disturbance entering the tank-4 state equation is

$$\xi_4 = \frac{\gamma_1 f_{p1}}{A_4}. \quad (35)$$

Importantly, the controller is not explicitly informed of the disturbance magnitude. The proposed DO-BSSTC estimates the disturbance from the measured plant response.

6 Performance Metrics

The tracking error for the controlled outputs is defined as

$$e_3 = h_3 - h_3^*, \quad e_4 = h_4 - h_4^*. \quad (36)$$

The integral absolute error is

$$IAE = \int_0^T |e(t)| dt. \quad (37)$$

The integral squared error is

$$ISE = \int_0^T e^2(t) dt. \quad (38)$$

The integral time absolute error is

$$ITAE = \int_0^T t|e(t)| dt. \quad (39)$$

To evaluate actuator smoothness, the control total variation is calculated as

$$TV = \sum_{k=1}^{N-1} |u(k+1) - u(k)|. \quad (40)$$

A smaller TV indicates a smoother control signal.

The RMS control effort is calculated according to

$$u_{\text{RMS}} = \sqrt{\frac{1}{N} \sum_{k=1}^N u^2(k)}. \quad (41)$$

For the disturbance experiment, the disturbance-window IAE is calculated over

$$700 \leq t \leq 900 \text{ s}. \quad (42)$$

The percentage improvement of DO-BSSTC relative to BSSTC is defined as

$$\text{Improvement} = \frac{J_{\text{BSSTC}} - J_{\text{DO-BSSTC}}}{J_{\text{BSSTC}}} \times 100\%. \quad (43)$$

7 Simulation Results

7.1 Nominal Performance

The nominal simulation results are summarized in Table 2.

Table 2: Nominal performance comparison.

Controller	IAE_{h3}	IAE_{h4}	ISE_{h3}	ISE_{h4}	TV
SMC	1112.94	767.76	20353.49	10214.87	4994050.00
BSSTC	1256.54	1122.00	20652.34	11052.39	364.36
DO-BSSTC	1256.54	1122.19	20652.28	11055.17	364.37

The corresponding RMS control efforts are

Table 3: Nominal RMS control effort.

Controller	RMS control effort
SMC	44.72%
BSSTC	40.48%
DO-BSSTC	40.48%

The nominal tracking results demonstrate that the proposed disturbance observer does not significantly alter the baseline BSSTC behavior.

The h_3 IAE is identical for BSSTC and DO-BSSTC at the reported numerical precision. The h_4 IAE changes from 1122.00 to 1122.19, corresponding to a difference of approximately -0.02% .

Similarly, the h_4 ISE changes from 11052.39 to 11055.17, corresponding to approximately -0.03% .

Therefore, the disturbance observer does not provide a nominal tracking improvement in this experiment. Its benefit becomes apparent when an unknown disturbance is introduced.

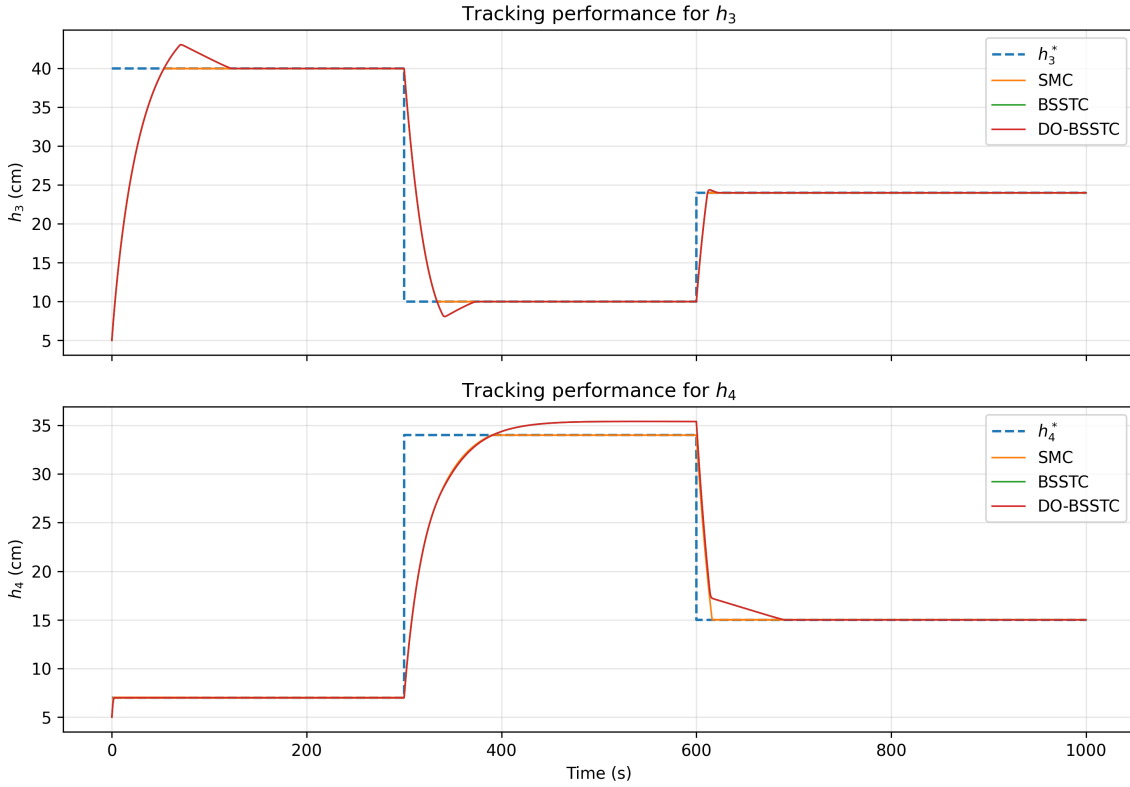


Figure 2: Liquid-level responses of the QTS for SMC, fixed-gain BSSTC, and proposed DO-BSSTC.

Figure 2 shows the liquid-level comparison between the three controllers. The controllers achieve regulation of the controlled tank levels for the different reference operating points.

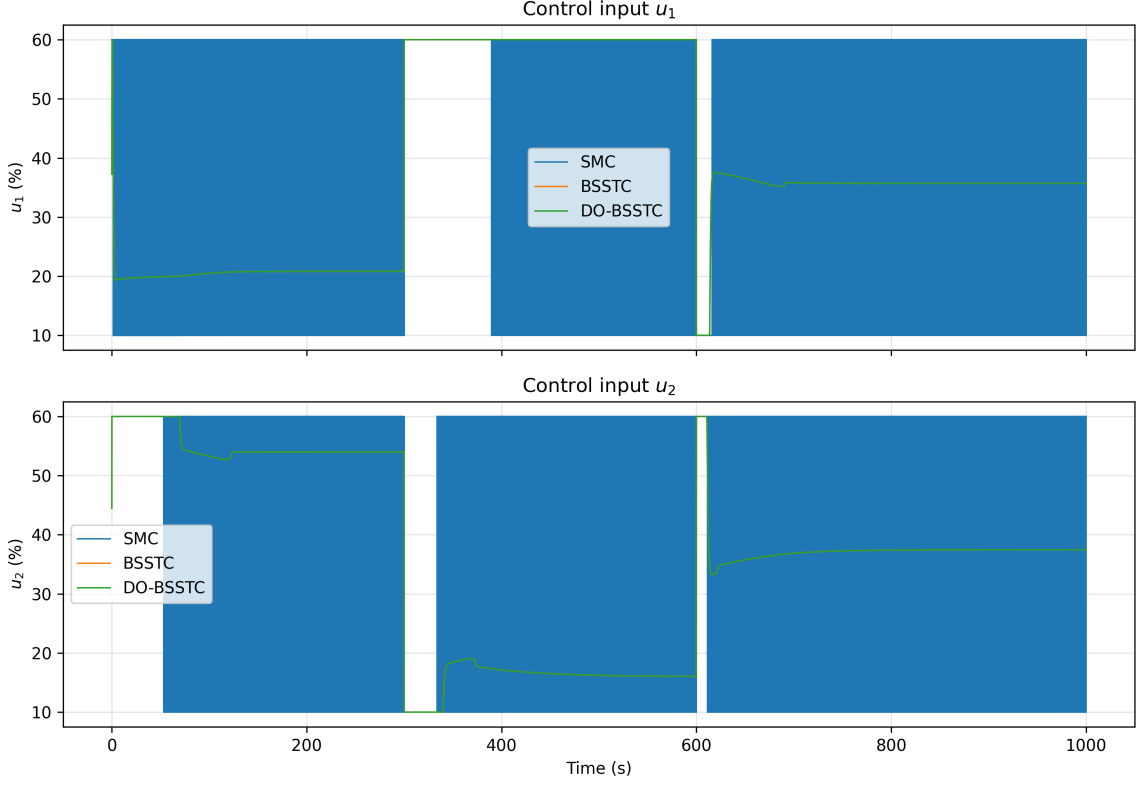


Figure 3: Comparison of pump control effort for SMC, BSSTC, and DO-BSSTC.

The control signals in Fig. 3 illustrate the substantial difference between first-order SMC and the super-twisting approaches.

The total variation of the SMC control signal is approximately

$$TV_{SMC} = 4.99405 \times 10^6, \quad (44)$$

whereas

$$TV_{BSSTC} = 364.36. \quad (45)$$

This corresponds to an approximate reduction of

$$99.993\% \quad (46)$$

in control total variation.

The result is consistent with the objective of super-twisting control: achieving robust control while producing a substantially smoother actuator signal than a conventional first-order relay SMC.

7.2 Disturbance Rejection

The most significant difference between BSSTC and DO-BSSTC appears during the disturbance experiment.

A pump-flow loss of

$$f_{p1} = -300 \text{ cm}^3/\text{s} \quad (47)$$

was introduced from 700 to 900 s.

The disturbance-window results are presented in Table 4.

Table 4: Tank-4 disturbance-rejection performance for $-300 \text{ cm}^3/\text{s}$ pump-flow loss.

Metric	BSSTC	DO-BSSTC	Improvement
Window IAE	27.56	0.40	98.55%
Mean $ e_4 $ (cm)	0.1378	0.0020	98.55%
Recovery time (s)	0.11	0.01	90.91%

The disturbance-window IAE decreases from

$$IAE_{\text{BSSTC}} = 27.56 \quad (48)$$

to

$$IAE_{\text{DO-BSSTC}} = 0.40. \quad (49)$$

Therefore,

$$\frac{27.56 - 0.40}{27.56} \times 100 = 98.55\%. \quad (50)$$

Thus, the proposed controller reduces the accumulated tank-4 tracking error during the disturbance interval by approximately 98.55%.

The mean absolute tracking error also decreases from

$$0.1378 \text{ cm} \quad (51)$$

to

$$0.0020 \text{ cm}. \quad (52)$$

This indicates that the proposed disturbance observer is able to compensate the investigated pump-flow loss before it produces a large persistent tracking deviation.

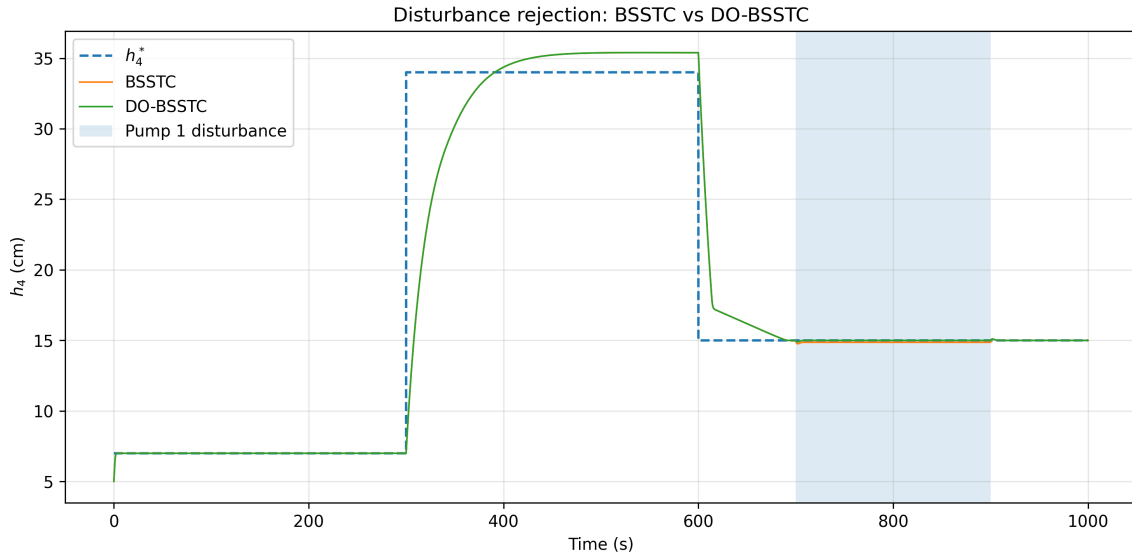


Figure 4: Disturbance-rejection experiment for the proposed DO-BSSTC compared with the fixed-gain BSSTC.

Figure 4 illustrates the tank-4 behavior during the disturbance interval. The fixed-gain BSSTC exhibits a measurable tracking deviation after the disturbance is introduced. In contrast, the proposed DO-BSSTC maintains the tank level much closer to its reference trajectory.

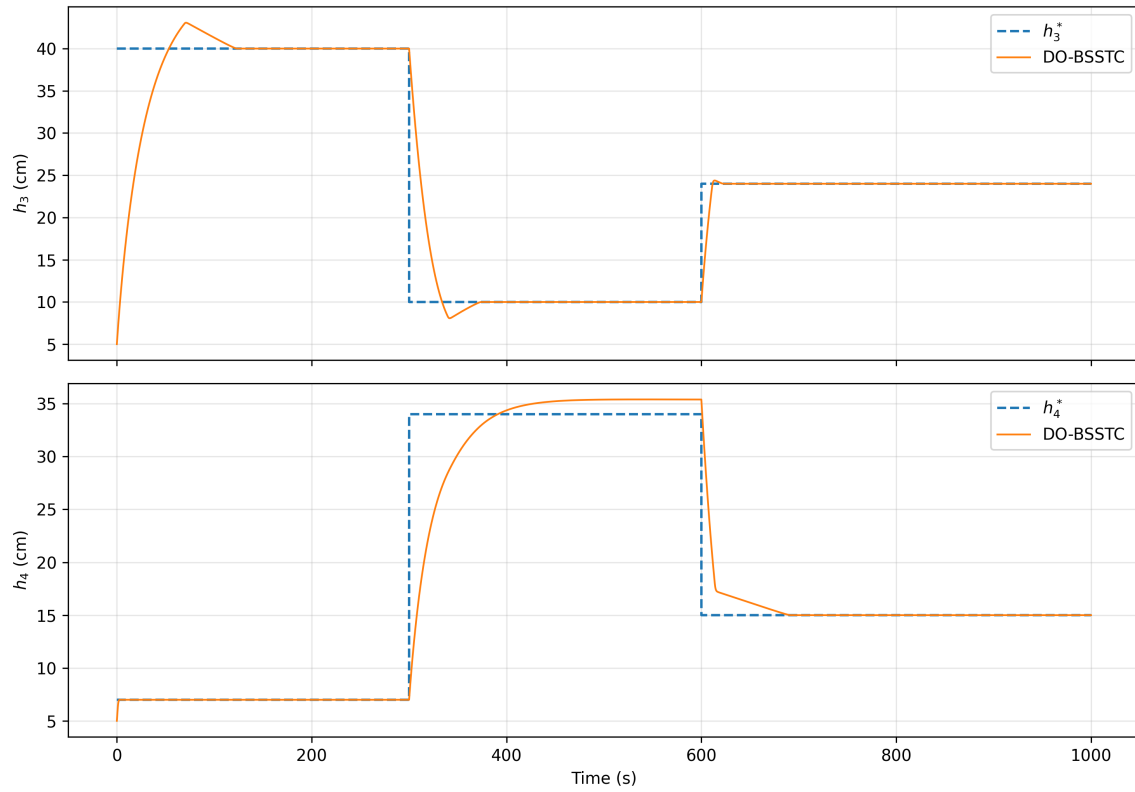


Figure 5: Liquid-level comparison between fixed-gain BSSTC and proposed DO-BSSTC during the disturbance experiment.

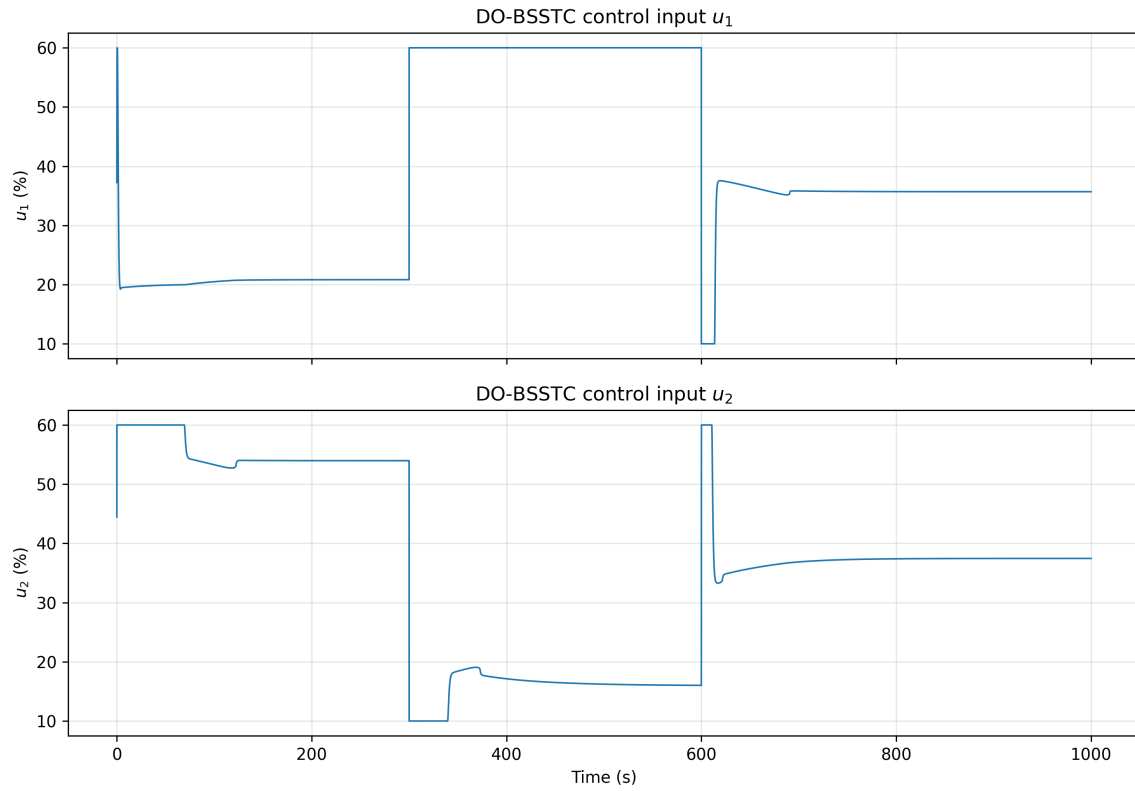


Figure 6: Pump control effort for BSSTC and DO-BSSTC.

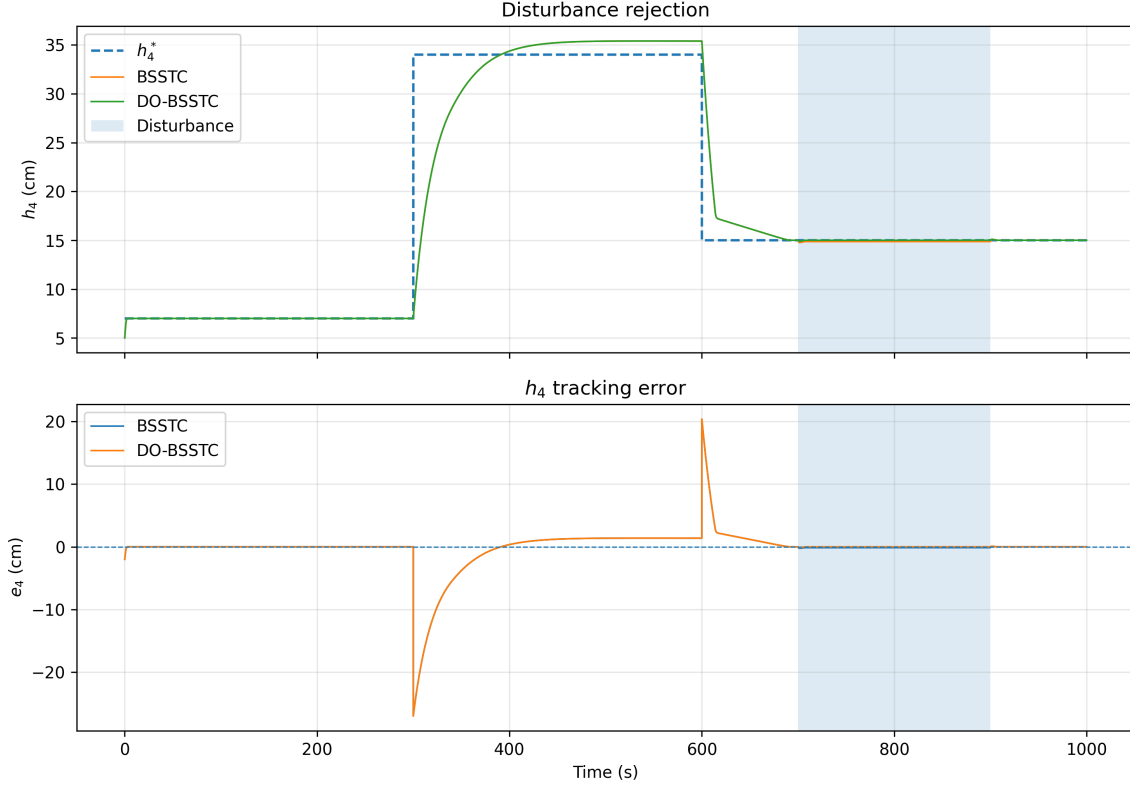


Figure 7: Detailed disturbance response and control action of the DO-BSSTC controller.

7.3 Post-Disturbance Recovery

The recovery time was evaluated after the disturbance was removed at $t = 900$ s. The criterion was

$$|e_4| < 0.1 \text{ cm.} \quad (53)$$

The fixed-gain BSSTC reached this criterion after approximately

$$t_{\text{rec,BSSTC}} = 0.11 \text{ s.} \quad (54)$$

The proposed DO-BSSTC reached the same criterion after

$$t_{\text{rec,DO-BSSTC}} = 0.01 \text{ s.} \quad (55)$$

The relative reduction is therefore

$$\frac{0.11 - 0.01}{0.11} \times 100 = 90.91\%. \quad (56)$$

It should be noted that this recovery metric is based on the first crossing of the specified 0.1 cm error threshold. Therefore, it should be interpreted as a threshold-based recovery indicator rather than a conventional settling-time definition.

7.4 Effect on Tank 3

The disturbance was introduced directly into the tank-4 equation. Consequently, the simulation results show essentially zero disturbance window error for tank 3.

The reported results were

Table 5: Disturbance-window results for tank 3.

Metric	BSSTC	DO-BSSTC	Difference
Window IAE	0.00	0.00	approximately -1.84%
Mean $ e_3 $	0.0000	0.0000	approximately zero

Because the absolute values are effectively zero, the percentage difference in this case is not physically meaningful. The important observation is that the investigated disturbance primarily affects the tank-4 controlled channel.

8 Discussion

The simulation results reveal an important distinction between nominal tracking performance and disturbance rejection.

Under nominal conditions, the proposed DO-BSSTC does not significantly outperform the fixed-gain BSSTC. The h3 tracking indices are essentially unchanged, while the h4 IAE and ISE differ by less than 0.03%.

This behavior is desirable because it demonstrates that the proposed disturbance observer does not introduce a significant degradation of the baseline controller.

The primary advantage is observed when the plant is subjected to the unknown pump-flow disturbance. The fixed-gain BSSTC must compensate for the disturbance through its feedback dynamics. The DO-BSSTC instead estimates the disturbance and generates a compensating feed-forward action.

The resulting reduction in disturbance-window IAE is substantial:

$$98.55\%. \quad (57)$$

The mean absolute error is similarly reduced by approximately two orders of magnitude.

Another important observation concerns actuator activity. The SMC controller has a control total variation of approximately 4.99×10^6 , while the BSSTC and DO-BSSTC have values near 364. This demonstrates that the continuous super-twisting structure provides a major actuator-smoothness advantage.

Therefore, the proposed architecture combines two useful properties:

$$\boxed{\text{robust continuous feedback} + \text{explicit disturbance compensation}} \quad (58)$$

The contribution should consequently not be interpreted as a replacement for the BSSTC under nominal conditions. Instead, DO-BSSTC extends the BSSTC by adding a mechanism specifically targeted at unknown disturbance rejection.

9 Limitations and Future Work

The present study is based on numerical simulations and uses a noise-free nonlinear plant model for the main comparison. Therefore, the results do not yet establish the behavior of the disturbance observer under realistic measurement noise, parameter uncertainty, sensor quantization, or unmodeled hydraulic dynamics.

Furthermore, only one disturbance magnitude was investigated:

$$f_{p1} = -300 \text{ cm}^3/\text{s}. \quad (59)$$

Future work should therefore investigate a wider range of disturbance magnitudes and profiles, including step disturbances, time-varying disturbances, and disturbances affecting different tanks.

A particularly important extension is the addition of measurement noise. Because the disturbance observer uses a derivative-related quantity, noise amplification can become an important practical issue. A noise-robust differentiator or modulating-function-based derivative estimator could therefore be combined with the proposed disturbance observer.

Additional experiments should also evaluate parameter uncertainty in A_i , a_i , γ_i , and pump gains k_i .

Finally, experimental validation using a physical quadruple-tank laboratory setup would be necessary to establish practical applicability.

10 Conclusion

A disturbance-observer-enhanced backstepping super-twisting controller (DO-BSSTC) was proposed for nonlinear liquid-level regulation of a quadruple-tank system.

The proposed controller extends a fixed-gain BSSTC by estimating an unknown disturbance from the difference between the measured plant dynamics and the nominal nonlinear model. The estimated disturbance is then converted into a feed-forward pump compensation.

Three control strategies were compared: conventional first-order SMC, fixed-gain BSSTC, and DO-BSSTC.

The nominal results show that DO-BSSTC preserves the behavior of the fixed-gain BSSTC. The h3 IAE is identical for both controllers, while the h4 IAE differs by only approximately 0.02%.

The most significant improvement occurs under the investigated $-300 \text{ cm}^3/\text{s}$ pump-flow disturbance. The tank-4 disturbance window IAE decreases from 27.56 for BSSTC to 0.40 for DO-BSSTC, corresponding to a 98.55% reduction. The mean absolute error decreases from 0.1378 cm to 0.0020 cm.

The threshold-based recovery time after disturbance removal also decreases from 0.11 s to 0.01 s.

In addition, the comparison with first-order SMC demonstrates the actuator-smoothness advantage of the super-twisting structure. The control total variation decreases from approximately 4.99×10^6 for SMC to approximately 364 for BSSTC and DO-BSSTC.

These results indicate that explicit disturbance estimation can substantially enhance the disturbance-rejection capability of a continuous nonlinear super-twisting controller while preserving its nominal behavior.

The next step toward experimental validation is to evaluate the proposed method under measurement noise, model uncertainty, multiple disturbance magnitudes, and physical actuator limitations.

References

- [1] I. Aranda-Cetraro, G. Pérez-Zúñiga, R. Rivas-Pérez, and J. Sotomayor-Moriano, “Nonlinear Robust Control by a Modulating-Function-Based Backstepping Super-Twisting Controller for a Quadruple Tank System,” *Sensors*, vol. 23, no. 11, p. 5222, 2023.
doi: 10.3390/s23115222.